Hyperpigmentation in Thyrotoxicosis

The occurrence of hyperpigmentation in thyrotoxic patients has been estimated to range from 2 percent to as high as 40 percent in large series. The increased cutaneous pigmentation can be localized or generalized and is more common in individuals with darker skin. The distribution often resembles that seen in Addison disease, with pigment deposition in the creases of the palms and soles. However, several distinguishing features exist: mucous membrane involvement is uncommon, and pigmentation of the nipples and genital skin is less pronounced.

Hyperpigmentation associated with thyrotoxicosis is thought to result from increased release of pituitary ACTH, which compensates for accelerated cortisol degradation. The response of hyperpigmentation to treatment for hyperthyroidism is generally poor. Autoimmune Graves disease has also been reported in association with vitiligo.

Melasma

Melasma is a common hypermelanosis that typically affects sun-exposed areas of the face. Its pathogenesis is not fully understood, but genetic and hormonal factors, in combination with UV radiation, play important roles. Specific precipitating factors include:

- Oral contraceptives
- Estrogen replacement therapy
- Mild ovarian or thyroid dysfunction
- Ovarian tumors
- Cosmetics
- Nutritional factors
- Phototoxic and photoallergic medications
- Antiepileptic drugs

Melasma is rarely reported before puberty and is significantly more common in women, especially those of reproductive age. Individuals with darker skin types (Fitzpatrick types IV, V, or VI) are more frequently affected.

Lesions appear as brownish macules with irregular borders and a symmetric facial distribution, often coalescing into a reticular pattern. Sun exposure exacerbates the condition. Three major patterns of distribution are recognized:

- **Centrofacial (63%)**: forehead, nose, chin, and upper lip
- **Malar (21%)**: cheeks and nose
- **Mandibular (16%)**: ramus of the mandible

The anterior chest and dorsal forearms may also be involved.

Under Wood's lamp examination, melasma is classically categorized as epidermal, dermal, or mixed. The epidermal type shows sharp borders and enhanced visibility under Wood's lamp, while the dermal or mixed type appears blotchy and is less apparent. In most cases, melanin is distributed in the basal layer of the epidermis and the dermis.

Sun exposure is a major contributing factor, and sun protection is essential in management. Epidermal pigmentation tends to respond better to topical treatments than dermal pigmentation. Effective hypopigmenting agents include:

- Hydroquinone
- Tretinoin cream
- Azelaic acid
- Rucinol
- Kojic acid

The Kligman formula—a combination of hydroquinone, tretinoin, and a weak topical corticosteroid—is widely used. Chemical peels and laser therapy may be beneficial but carry a risk of inducing further hyperpigmentation. In some cases, melasma gradually resolves after withdrawal of hormonal triggers and with strict sun avoidance.

Pigmentation Changes in Pregnancy

During pregnancy, increased pigmentation occurs in approximately 90% of women and is more pronounced in those with darker skin types. Pre-existing pigmented lesions such as nevi and ephelides often darken, and recent scars may also become hyperpigmented. Normally pigmented areas—including the nipples, areolae, and genitalia—typically exhibit intensified coloration.

The linea alba, the midline of the anterior abdominal wall, frequently becomes hyperpigmented during pregnancy and is then referred to as **linea nigra**. A small proportion of pregnant women develop hyperpigmentation in the axillae or inner upper thighs.

Melasma, commonly known as the "mask of pregnancy," affects more than 50% of pregnant women.

Acanthosis Nigricans

Acanthosis nigricans is a condition characterized by hyperpigmentation and velvety thickening of the skin, typically in intertriginous areas. It is associated with insulin resistance, obesity, and endocrinopathies, and is discussed in detail in Chapters 152 and 154.

Diabetes and Skin Pigmentation

(See Chapter on Diabetes)